Tanmay Dipak Kowlgi
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EDUCATION

Purdue University, West Lafayette, IN **Aug 2023 - May 2025**
Master of Science in Mechanical Engineering **GPA: 3.84/4**
Relevant Coursework & Projects: Control Systems, Microprocessors, Robotics, Mechatronics, Industrial IOT, Design for Manufacturability
Savitribai Phule Pune University, Pune, India **Jun 2016 - Apr 2020**
Bachelor of Mechanical Engineering **GPA: 9.25/10**

SKILLS

Mechatronics: Simulink, Raspberry Pi, Arduino, Interfacing sensors & motors with microcontrollers, Circuit design, Microprocessors
Robotics: Operating robotic grippers, path planning, robotic simulation software (Rviz), ROS
Programming Languages: MATLAB, Java, Python, SQL, C++, C, Spring boot, Seeburger BIS
AI, ML & other Software skills: Exploratory data analysis, tuning & training AI models, AWS, Agile methodology – JIRA, Microsoft Office
Mechanical Design: SolidWorks, CREO, Onshape, 3D printing, Manufacturing, GD&T, Ansys Granta EduPack**,** Machining – lathe, drilling, cutting, grinding
Domain: Banking, Finance, Manufacturing

WORK EXPERIENCE

TSMC Arizona Corporation, Phoenix, AZ
Equipment Engineer (Full-Time) **Jun 2025 - Present**

- Maintained and optimized complex **mechatronics & robotic** systems operating under high-vacuum, high-temperature conditions with sub-micron precision requirements for **4nm/5nm chip** manufacturing processes.
- Diagnosed and resolved failures in automated process systems using structured root cause analysis.
- Conducted structured root cause analysis on a critical equipment failure, diagnosed a targeted component issue and saving $40K which avoided replacement costs.
- Reduced unplanned tool-downtime by 20% through revised PM processes.

Epitaxy (EPI) Equipment Engineer Intern **May 2024 – Aug 2024**

- Enhanced accuracy of sensor data collection by 50% by proposing a solution minimizing false alarms in Silicon wafer scratch prevention system.
- Devised a strategy to reduce tool downtime by 4 hours and minimize engineering resource downtime, optimizing the workload of 2 engineers.
- Identified and resolved a critical bug in Silicon wafer production system, preventing scratches and saving company millions of dollars annually.

Rapid Prototyping Lab, Purdue University, West Lafayette, IN **Oct 2023 – Apr 2025**
Teaching Assistant

- Provided expert guidance to over 100 students each semester in 3D printing techniques, advising them on material properties, machine settings, and print pattern optimization customized for their design projects.
- Maintained and managed 3D printing machines, resolving students' inquiries and performing repairs to ensure continuous, uninterrupted operation.

Alphaniti Fintech Pvt. Ltd., Pune, India **Aug 2022 - Jun 2023**
Data Analyst

- Augmented existing stock data models by 5% by making use of data collection, data mining, & data analysis with Python, SQL & MongoDB.
- Automated 3000 Investment rationale reports with Linux & Python.
- Formulated strategy with CEO, CTO & cross functional teams to translate crucial business requirements such as automation of Management Information System (MIS) into actionable products & productivity improvement.
- Reduced existing API call time from 180 seconds to 30 seconds by algorithm optimization in Python.

Infosys Limited, Pune, India **Nov 2020 - Aug 2022**
Senior Systems Engineer

- Minimized business risk of client by leading resolution of critical production issues, preventing loss of 25000 AUD, ensuring seamless operations.
- Served as a Seeburger BIS (Java based language) developer by resolving 20+ new features & bug fixes for global banking clients in B2B domain.
- Constructed a robust website with Java, Springboot, microservices, SQL & AWS enabling customer profile creation & product display.

Fenss Technologies Pvt. Ltd., Pune, India **Apr 2019 - Oct 2019**
Intern

- Spearheaded integration of 10+ sensors & actuators with ARM microcontrollers for an Automated medicine Dispensing System (ADS).
- Developed electrical circuits and mechatronic system code with Python programming language.
- Created object detection algorithm for detecting packaging & conducted unit/integration testing of ADS subsystems.
- Led vendor interactions and provided exceptional customer support.

RESEARCH EXPERIENCE

Mechanisms and Robotics Systems (MARS) Lab, Purdue University, West Lafayette, IN **Aug 2023 - Dec 2023**
Independent Study – Robotics with Tactile sensing

- Integrated imitation learning for robotic manipulation with tactile sensing & fixture design.
- Designed & manufactured robotic grippers with gelsight mini (3D Tactile sensing device) end effector using Onshape.
- Collaborated with PhD students & carried out robotic simulation (Rviz), hardware testing & object-oriented programming in Python & ROS.

ENGINEERING PROJECTS

Redesigned Upper Wishbone using Design for Manufacturability principles **Feb 2025 – Apr 2025**

- Redesigned Upper Wishbone using DFM principles & improved manufacturability by eliminating piping structures, reduced the weight of the component by topology optimization and welding cost by **24%** per component

Mechatronics Project: Autonomous bumble bee robot **Jan 2024 - Apr 2024**

- Built an Arduino-powered autonomous robot capable of detecting, sorting, & delivering color-coded pollen using integrated sensors, PID control, & servos.

AI model for Kirby Risk Industry **Jan 2024 - Apr 2024**

- Pioneered an AI ANN autoencoder model to distinguish between scrap and non-scrap components using sound, position and acceleration data points of the 4-axis milling machine motors to reduce physical quality control process.

Designed a Control System Model for an engine **Nov 2023 - Dec 2023**

- Executed PI closed loop system model to regulate speed of an engine operated by pneumatic valve leveraging Simulink, STM32 & C.

Constructed a Control System Model of a DC Motor & calculated speed of motor by applying interrupt logic **Nov 2023**

- Simulated PI model of DC motor to reject disturbances in Simulink and verified model with STM32, C.
- Implemented an interrupt service software to measure speed operating with optical sensor.

Low-cost obturation device for RCT using Heat and Vibration **Jun 2019 - Feb 2020**

- Optimized an obturation device's vibration frequency and temperature via Arduino.
- Designed sensor holding fixtures in CREO & 3D printed with ABS thermoplastic.

Robot path planning using PRM (probabilistic roadmap) **Feb 2019**

- Created algorithm using MATLAB & VREP enabling a virtual mobile robot to compute an optimized path between 2 points on an obstacle arena.

CERTIFICATION

- **PG Certification in Data Science**, IIIT Bangalore (2021) – Python, SQL, ML, Time Series, PCA, Regression, Decision Trees
- **Coursera Robotics Specialization** – Robot Motion, Kinematics, Dynamics, Motion Planning (2018–2019)
- **Coursera IoT Specialization** – Embedded Systems, Arduino Programming, Sensor Interfacing (2018–2019)
- **Software Development Courses (Coursera):** Computer Vision Basics, Programming Foundations with JavaScript, HTML & CSS (Apr - Jun 2020)
- **Creo CAD Modeling Certification** – Grade A (2017)

PUBLICATION

Challenges in the Development of a New Obturation Device: A Critical Survey in Pune City [ISBN: 978-93-89107-65-4], presented at the Institute for Engineering Research and Publication [IFERP] conference held on 12th & 13th December 2019 at Nagpur, Maharashtra, India.